Privacy-preserving Non-negative Matrix Factorization with Outliers

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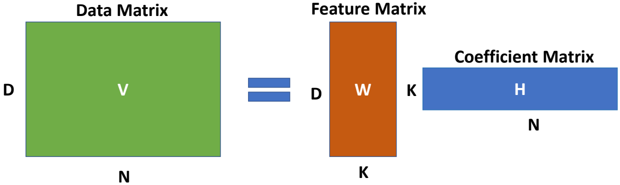
What is Non-negative Matrix Factorization (NMF)?



- Decompose non-negative matrix **V** into two lower rank ($K \ll D$) non-negative matrices - **W** and **H**.

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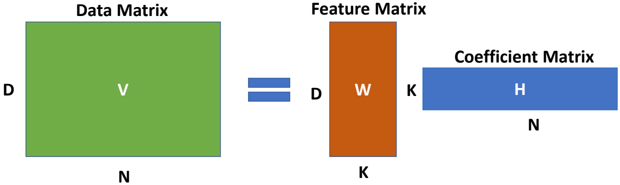

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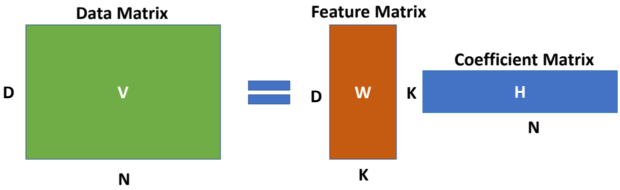
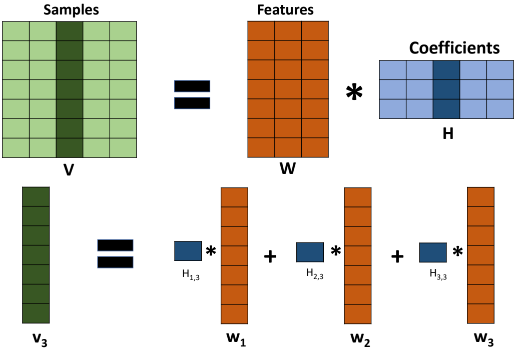
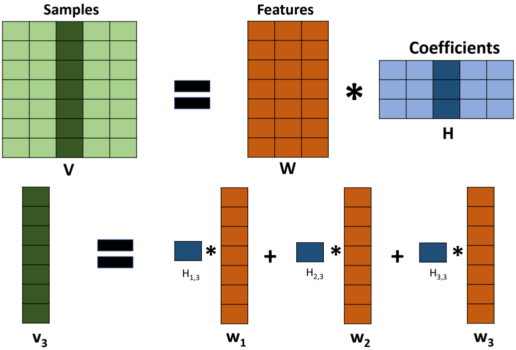


- Decompose non-negative matrix **V** into two lower rank ($K \ll D$) non-negative matrices - **W** and **H**.
- **W** $\rightarrow$ Capture useful feature information of population.
- **H** $\rightarrow$ Store user specific information.

NMF Interpretation

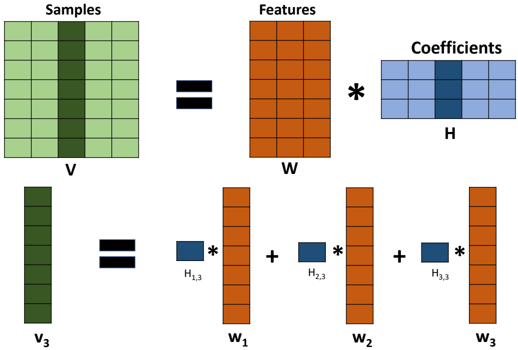


- *i*-th column vector of **V** $\rightarrow$ linear combination of columns of **W**.



- i-th column vector of $\mathbf{V} \rightarrow$ linear combination of columns of $\mathbf{W}$.
- i-th column vector of $\mathbf{H} \rightarrow$ supply coefficients.

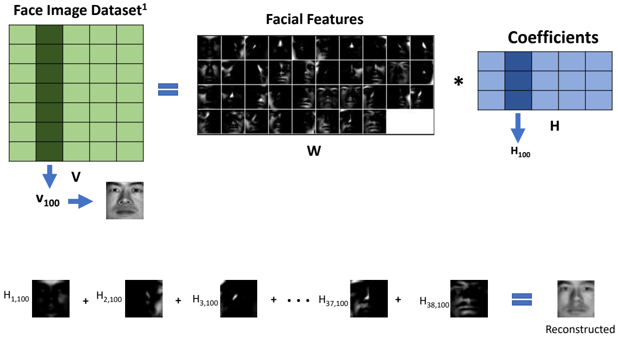
NMF Interpretation



- i-th column vector of $\mathbf{V} \rightarrow$ linear combination of columns of $\mathbf{W}$.
- i-th column vector of $\mathbf{H} \rightarrow$ supply coefficients.
- Application $\rightarrow$ image segmentation, topic modeling, dimension reduction, community detection in social networks.

Application of NMF

Extracting local facial feature.



- Individual column vector of **V** $\rightarrow$ individual pixel information.
- **W** $\rightarrow$ stores facial feature.

[1] The Extended Yale Face Database B

Topic 1
sport
Football
NFL
players

Topic 2
President
Donald Trump
Hillary Clinton
election

LeBron James says President Trump 'trying to divide through sport'

Basketball star LeBron James has praised the American football players who have protested against Donald Trump, and accused the US president of "using sports to try and divide us".

Trump said on Friday that NFL players who fail to stand during the national anthem should be sacked or suspended. In widespread protests at the weekend, players responded by kneeling, linking arms or staying in the locker room.

James praised the players' unity, and said: "The people run this country."

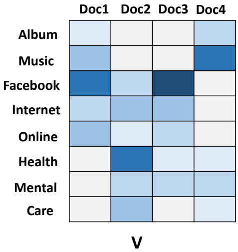
The 32-year-old added: "I'm not going to let one individual, no matter their power, ever use sport as a platform to divide us.

"Sport is so amazing, what it can do for everyone. No matter the shape, size, weight, ethnicity, religion or whatever - people find teams, players and colours because of sport. It brings people together like none other."

James, who plays for the Cleveland Cavaliers and has won three NBA championships, campaigned for Hillary Clinton, Trump's rival, during the 2016 presidential election campaign [2].

- Topic Modeling → Identify topic(s) of a document.

[2] https://www.bbc.com/sport/american-football/41393513



=

	Topic1	Topic2	Topic3
Album	3	0.1	0.2
Music	2.5	0.3	0.1
Facebook	0.2	2.7	0.7
Internet	0.1	2.6	0.4
Online	0.3	2.8	0.2
Health	0.8	0.1	3
Mental	0.2	0.6	3.2
Care	0.6	0.3	3.8

W

*

1.1	0.5	0.2	1.8
1.5	1.3	1.4	0.1
0.2	1.2	1	1.1

H

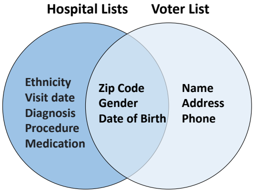
- Three topics $\rightarrow$ Entertainment, Online activity and Health issue.
- **H** $\rightarrow$ How much a document is related to certain topic.

Privacy Leakage in NMF

V and H contain participant's sensitive data.
Information leakage can violate privacy of participant.

We focus on development of privacy- preserving NMF framework to learn $\mathbf{W}_{private}$.

Why do we need privacy?



- Auxiliary information → Anonymization fails to provide privacy.[3]

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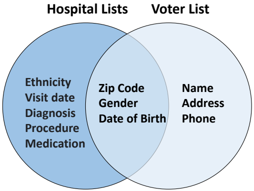

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- Break Anonymity of the Netflix Prize Data set.[4]

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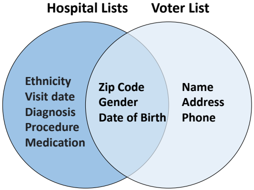

- Auxiliary information → Anonymization fails to provide privacy.[3]
- Break Anonymity of the Netflix Prize Data set.[4]
- Personal information leakage → main hindrance to collecting and analysing sensor data.

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Online Nonnegative Matrix Factorization With Outliers[5]

NMF decomposition:

$$\mathbf{V} \approx \mathbf{W}\mathbf{H} + \mathbf{R}$$

- $\mathbf{R} \rightarrow$ Outlier Matrix.

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Objective function:

$$\min_{\mathbf{W}\in\mathcal{C},\mathbf{H}\geqslant 0,\mathbf{R}\in\mathcal{Q}} \frac{1}{2}\left\|\mathbf{V}-\mathbf{WH}-\mathbf{R}\right\|_2^2 + \lambda\left\|\mathbf{R}\right\|_1$$

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Loss function:

$$J = \min_{\mathbf{W}\in\mathcal{C},\mathbf{H}\geqslant 0,\mathbf{R}\in\mathcal{Q}} \frac{1}{2}\left\|\mathbf{V}-\mathbf{WH}-\mathbf{R}\right\|_2^2 + \lambda\left\|\mathbf{R}\right\|_1$$

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- Projected Gradient Descent Method.

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- $\mathbf{H}_{t+1} = \mathcal{P}_+(\mathbf{H}_t - \eta_H \bigtriangledown_H J(\mathbf{V}, \mathbf{W}_t, \mathbf{H}_t, \mathbf{R}_t))$.

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- $\mathbf{R}_{t+1} = S_{\lambda,M}\left(\mathbf{V}-\mathbf{W_t}\mathbf{H_{t+1}}\right)$
- $\mathbf{W}_{t+1} = \mathcal{P}_{\mathcal{C}}(\mathbf{W}_t - \eta_W \bigtriangledown_W J(\mathbf{V},\mathbf{W}_t,\mathbf{H}_{t+1},\mathbf{R}_{t+1}))$

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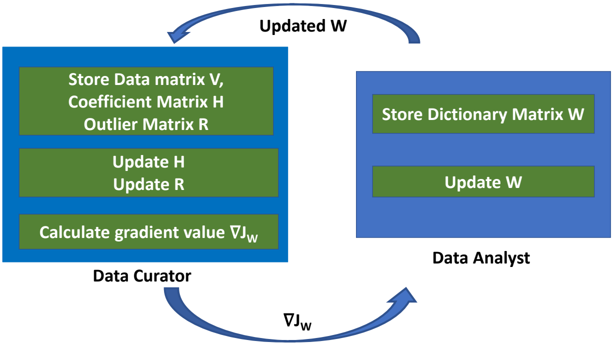
Privacy preserving framework of NMF



- Separate private and non-private training node.

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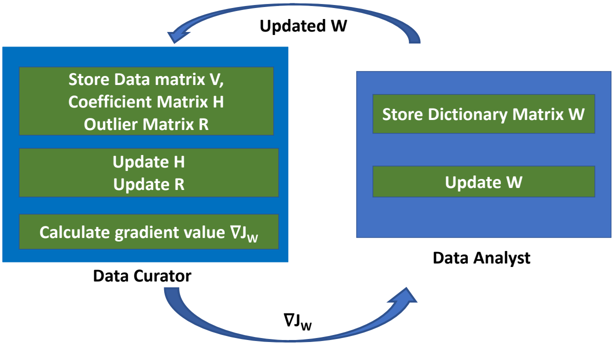
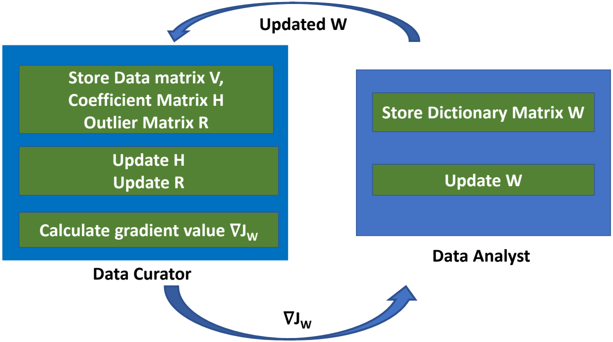
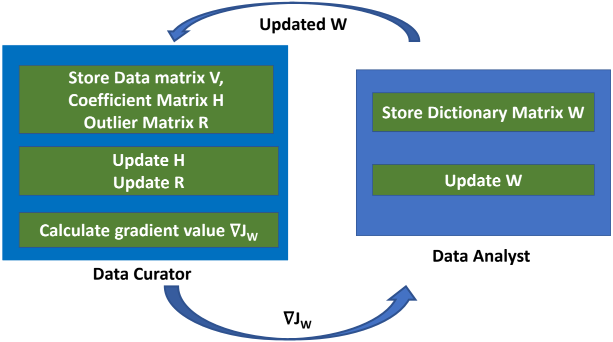


- Separate private and non-private training node.
- Two nodes $\rightarrow$ Data Curator & Data Analyst.

Privacy preserving framework of NMF

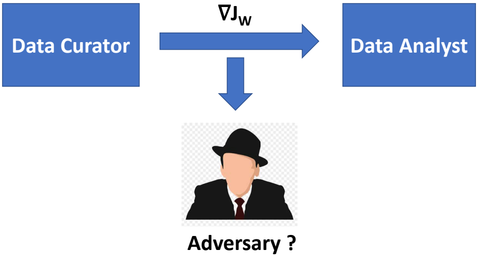


- Separate private and non-private training node.
- Two nodes → Data Curator & Data Analyst.
- Data Curator → Store and train private data.



- Separate private and non-private training node.
- Two nodes $\rightarrow$ Data Curator & Data Analyst.
- Data Curator $\rightarrow$ Store and train private data.
- Data analyst $\rightarrow$ Store and train non-private data.

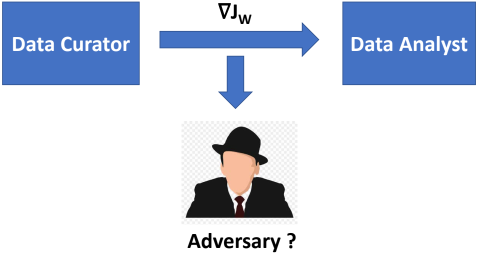
Publicly Shared Gradients Free from Privacy-Attack?



- Leakage from Gradients → Sharing the gradients can leak private training data.[6]

[6] Zhu, L., Liu, Z. and Han, S., 2019. Deep leakage from gradients. Advances in Neural Information Processing Systems, 32.

Publicly Shared Gradients Free from Privacy-Attack?



- Leakage from Gradients $\rightarrow$ Sharing the gradients can leak private training data.[6]
- Dummy gradient from dummy input data $\rightarrow$ Reconstruct private training data.

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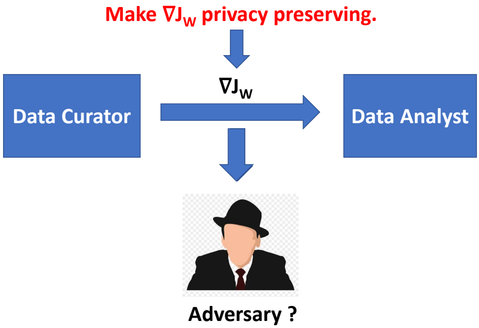
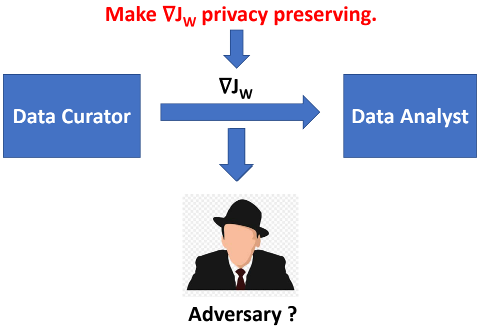


- $\nabla_W J() \rightarrow$ Make privacy preserving.

Our proposed solution



- $\nabla_W J() \rightarrow$ Make privacy preserving.
- Solution $\rightarrow$ Differential Privacy (DP)

Dose Bob has cancer?

- Data set D $\rightarrow$ Model $\rightarrow$ 0.53

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- Data set D+Bob $\rightarrow$ Model $\rightarrow$ 0.55

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Maybe Bob has cancer?

Intuition behind Differential Privacy

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- Data set D+Bob $\rightarrow$ Model $\rightarrow$ ~~0.55~~ 0.90

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Bob has cancer!
Privacy Leak!

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Maybe Bob has cancer?

- Data set D+Bob $\rightarrow$ Model $\rightarrow$ ~~0.55~~ 0.90

Bob has cancer!
Privacy Leak!

Privacy loss: $\log(\frac{0.55}{0.53}) = 0.037$
privacy loss: $\log(\frac{0.90}{0.53}) = 0.53 \rightarrow$ 14% times privacy loss !

Differential Privacy limits the upper bound of privacy loss.

DP: Definition and Mathematical Expression

Privacy loss for mechanism f:

$$\log \frac{P(f(D) \in \mathcal{S})}{P(f(D') \in \mathcal{S}}) \leqslant \epsilon$$

- Mechanism f() $\rightarrow$ Any kind of computation that uses sensitive data.
- D and D' $\rightarrow$ Neighbouring Data set.
- $\epsilon \rightarrow$ Privacy budget.

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(ϵ, δ)-Differential Privacy for mechanism f:

$$P(f(D) \in \mathcal{S}) \leqslant \delta + e^{\epsilon} P(f(D') \in \mathcal{S})$$

ϵ-DP mechanism guarantees its privacy with probability $1 - \delta$.

Definition: (ϵ, δ-DP [7])

An algorithm $f : \mathcal{D} \mapsto \mathcal{T}$ provides (ϵ, δ)- differential privacy if $P(f(D) \in \mathcal{S}) \leqslant \delta + e^{\epsilon} P(f(D') \in \mathcal{S})$ for all measurable $\mathcal{S} \subseteq \mathcal{T}$ and for all neighbouring data sets $D, D' \in \mathcal{D}$

[7] Dwork, C., McSherry, F., Nissim, K. and Smith, A., 2006, March. Calibrating noise to sensitivity in private data analysis. In Theory of cryptography conference (pp. 265-284). Springer, Berlin, Heidelberg.

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- Lower $(\epsilon, \delta) \rightarrow$ Higher privacy guarantee

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But how differential privacy guarantee privacy ?

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Adding Noise or Randomness

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Randomness and Privacy

- Blurry Image $\rightarrow$ Increase Privacy

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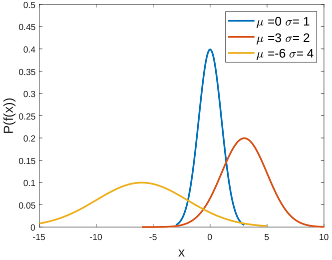
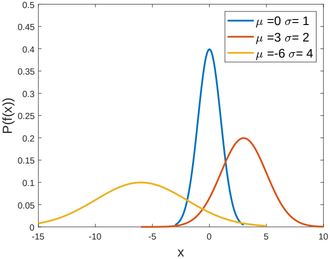


- Adding noise $\rightarrow$ Smooth over possible outcome.

- Blurry Image $\rightarrow$ Increase Privacy



- Adding noise $\rightarrow$ Smooth over possible outcome.

We use Gaussian Mechanism to introduce randomness

$\mathcal{L}_2$ sensitivity [8] :

$$\Delta_2 := \max_{D,D'} \|f(D) - f(D')\|_2$$

- Sensitivity $\Delta_2 \rightarrow$ Max output change by a single participant.

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Adding noise to preserve privacy: Sensitivity

$\mathcal{L}_2$ sensitivity [8] :

$$\Delta_2 := \max_{D,D'} \|f(D) - f(D')\|_2$$

- Sensitivity $\Delta_2 \rightarrow$ Max output change by a single participant.
- High Sensitivity $\Delta_2 \rightarrow$ more noise to meet the privacy guarantee.

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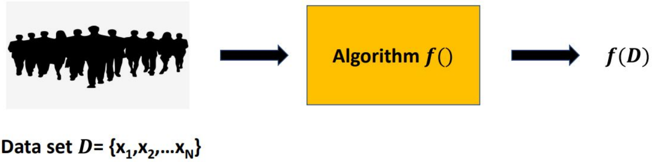


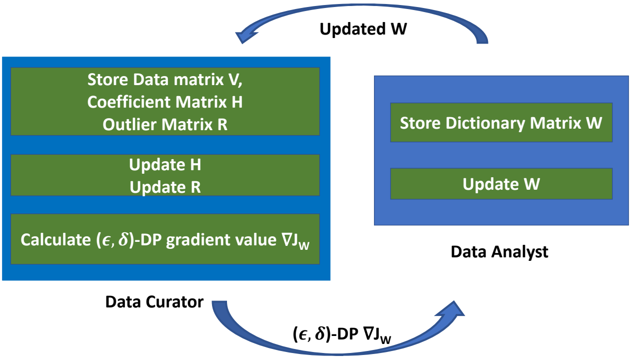
Gaussian Mechanism [9]:

$$\text{Releasing } f(D) + S \text{ where } S \sim \mathcal{N}(0, \tau^2)$$

$$\tau = \frac{\Delta_2}{\epsilon}\sqrt{2\log\frac{1.25}{\delta}}$$

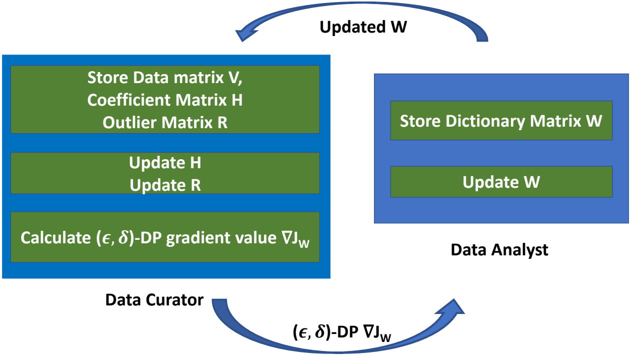
Algorithm f satisfies (ϵ, δ)- Differential privacy.

[9] Dwork, C. and Roth, A., 2014. The algorithmic foundations of differential privacy. Found. Trends Theor. Comput. Sci., 9(3-4), pp.211-407.



- Calculate sensitivity of $\nabla_W J$.
- Select privacy budget (ϵ, δ).
- (ϵ, δ)-DP $\nabla_W J$

(ϵ, δ)-DP MMF



- Calculate sensitivity of $\nabla_W J$.
- Select privacy budget (ϵ, δ).
- (ϵ, δ)-DP $\nabla_W J$
- Release $\mathbf{W}_{private}$ for further research works.

Experimental Result

Guardian News Articles
Uci-news-aggregator
RCV1
TDT2

Text Data set

YaleB
CBCL

Face Images Data set

- 6 real data sets $\rightarrow$ 4 text data sets, 2 face image data sets .

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- Performance indices:

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CBCL

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- 6 real data sets $\rightarrow$ 4 text data sets, 2 face image data sets .
- Performance indices:
  - Text Data set $\rightarrow$ Topic word distribution.

Experimental Result

Guardian News Articles
Uci-news-aggregator
RCV1
TDT2

Text Data set

YaleB
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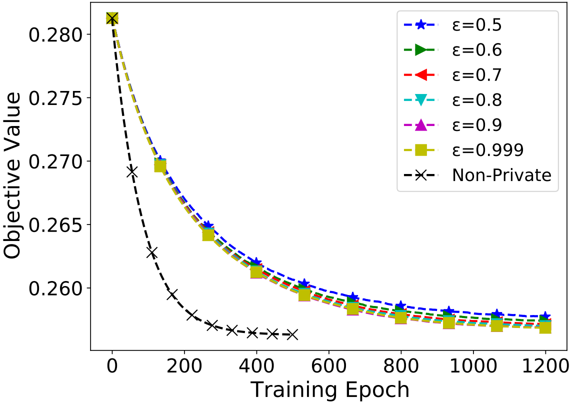
$$\text{Objective Value} = \frac{1}{2N} \|\mathbf{V} - \mathbf{WH}\|_2^2$$

- Hyper parameter selection $\rightarrow$ Follow the original study except the step size. $\eta_W \ll \eta_H$

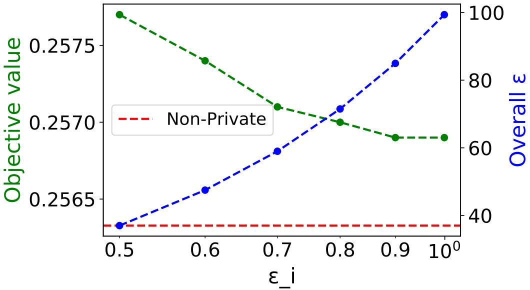
News aggregator Data set [10]



- Privacy preserving $\rightarrow$ need more epochs to converge.

[10] Dheeru Dua and Casey Graff. UCI Machine Learning Repository. 2017. url: http://archive.ics.uci.edu/ml

News aggregator Data set [10]



- Little utility gap.

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[11]Ilya Mironov. Rényi differential privacy. In: 2017 IEEE 30th Computer Security Foundations Symposium (CSF). IEEE. 2017, pp. 263275

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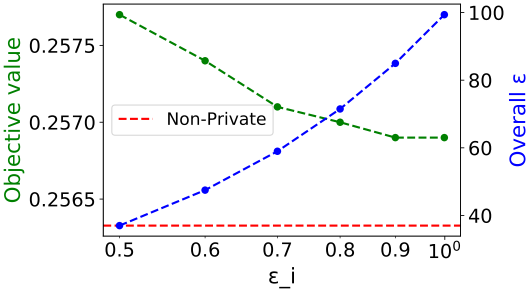

- Little utility gap.
- RDP $\rightarrow$ overall ϵ in the multi-stage composition[11].

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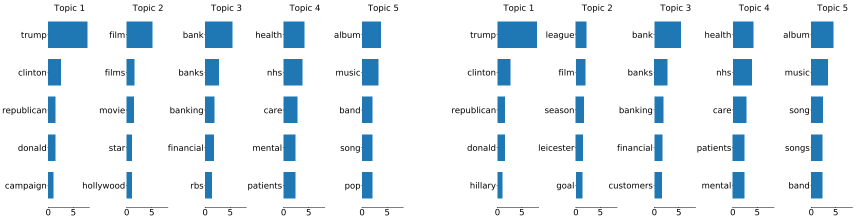

(a) Non-private (b) $(\epsilon = 0.5, \delta = 10^{-5})$-DP private

- The topic words distribution is almost similar.

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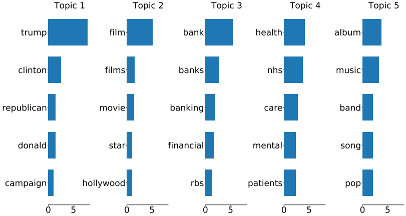
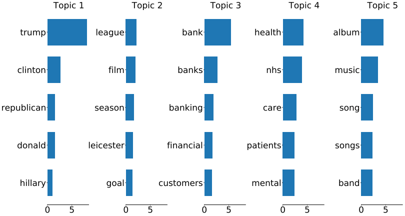


(a) Non-private



(b) $(\epsilon = 0.5, \delta = 10^{-5})$-DP private

- The topic words distribution is almost similar.
- Topic Coherence Score:

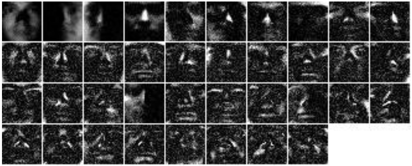
Non-private: 0.4658

$(\epsilon = 0.5, \delta = 10^{-5})$-DP: 0.451

[10] Dheeru Dua and Casey Graff. UCI Machine Learning Repository. 2017. url: http://archive.ics.uci.edu/ml

Yaleb[12]

(a) Non-private



(b) $(\epsilon = 0.5, \delta = 10^{-5})$-DP private

- $(\epsilon = 0.5, \delta = 10^{-5})$-DP $\rightarrow$ Little difference with non-private.

[12] The Extended Yale Face Database B. http://vision.ucsd.edu/ leekc/ ExtYaleDatabase/ExtYaleB.html

Summary

The proposed novel privacy-preserving NMF algorithm:

- It benefits from (ϵ, δ)-DP and has performance close to that of the non-private algorithm.

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- It has control to select privacy budget (ϵ, δ) based on the utility gap and overall ϵ.

The proposed novel privacy-preserving NMF algorithm:

- It benefits from (ϵ, δ)-DP and has performance close to that of the non-private algorithm.
- It has control to select privacy budget (ϵ, δ) based on the utility gap and overall ϵ.
- It can implement in any NMF-related task.

Question?

Thank you